HotelRevAI
AI-Driven Revenue Analysis for Hotels

Final Internship Project Report

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ABSTRACT

The hospitality industry generates a large amount of data through hotel reservations, guest records, and revenue transactions. Analyzing this data is essential for understanding business performance and improving decision-making. This project focuses on analyzing hotel booking and revenue data from **2018 to 2020** using **Microsoft Power BI**. The study aims to identify important trends related to revenue generation, booking patterns, customer behavior, and operational performance in hotels.

The dataset used in this analysis includes information such as hotel type, booking channels, customer types, number of guests, length of stay, and special requests made by customers. Power BI was used as the main tool to clean, transform, and visualize the data through interactive dashboards. Key performance indicators such as **Total Revenue, Average Daily Rate (ADR), Total Bookings, and Average Length of Stay** were analyzed to understand the financial and operational performance of the hotels.

The results of the analysis show that online travel agencies contribute a significant share of bookings, while most customers are first-time guests who prefer short stays. The study also identifies seasonal trends and differences in revenue generation between weekdays and weekends. The insights obtained from this project demonstrate how data analytics and visualization tools can help hotel management make better strategic decisions, improve customer satisfaction, and increase overall revenue performance.

Chapter 1

INTRODUCTION

The hospitality industry is one of the fastest-growing sectors of the global economy. It includes businesses that provide services related to accommodation, tourism, food, and entertainment. Hotels, resorts, restaurants, and travel agencies are key components of this industry. Among these, hotels play a vital role by offering lodging facilities to tourists, business travelers, and other visitors.

With the rapid growth of global tourism, the demand for hotel services has increased significantly. According to tourism studies, millions of travelers book hotel accommodations every year for both leisure and business purposes. As a result, the hotel industry has become highly competitive. Hotels must constantly improve their services, optimize pricing strategies, and enhance customer experience to remain competitive in the market.

In the past, hotel management relied heavily on manual reporting systems and basic financial statements to track their business performance. However, these traditional methods were often time-consuming and lacked the ability to provide deep insights into customer behavior and operational efficiency.

In recent years, technological advancements have introduced **Business Intelligence (BI) tools** that allow businesses to analyze large datasets and generate valuable insights. These tools help organizations make informed decisions based on real-time data analysis.

One of the most widely used BI tools is **Power BI**, developed by Microsoft. Power BI allows organizations to create interactive dashboards, visualizations, and reports that simplify complex datasets and help managers understand business trends.

In the hospitality industry, data analytics has become a powerful tool for improving revenue management, customer satisfaction, and operational performance. By analyzing booking data, hotels can identify trends such as seasonal demand, preferred booking channels, and customer preferences.

This project focuses on analyzing **hotel booking and revenue data from 2018 to 2020** using Power BI. The analysis aims to identify patterns in hotel bookings, revenue generation, market segments, and customer behavior.

1.1 Background of the Study

Hotel management involves monitoring several key performance indicators (KPIs) that help evaluate the overall success of the business. Some of the most important KPIs in the hotel industry include:

- Total Revenue
- Average Daily Rate (ADR)
- Total Bookings
- Occupancy Rate
- Average Length of Stay
- Customer Retention Rate

These metrics help hotel managers measure financial performance and operational efficiency.

The dataset used in this project contains hotel booking information from **July 2018 to August 2020**. It includes data related to two types of hotels:

- **City Hotel**
- **Resort Hotel**

The dataset also includes details about:

- Booking dates
- Market segments
- Customer types
- Length of stay
- Special requests
- Revenue generated
- Number of guests

According to the dashboard created using Power BI, the following key statistics were identified:

- **Total Revenue Generated:** 30.5 Million
- **Total Bookings:** 86.8K
- **Total Guests:** 168.7K
- **Average Daily Rate (ADR):** 100.8
- **Average Length of Stay:** 3.5 days

These metrics provide an overall view of the hotel's business performance during the study period.

The analysis also highlights how revenue and bookings vary based on different factors such as:

- Market segments
- Customer types
- Hotel type
- Seasonal trends
- Booking behavior

Understanding these patterns can help hotel managers make better decisions related to pricing, marketing strategies, and resource allocation.

1.2 Importance of Data Analytics in the Hotel Industry

Data analytics plays an essential role in modern hotel management. Hotels collect vast amounts of data every day through reservation systems, online booking platforms, customer feedback, and financial transactions.

If this data is properly analyzed, it can provide valuable insights that help hotels improve their performance and profitability.

Some major benefits of data analytics in the hotel industry include:

1. Revenue Optimization

Hotels can analyze historical booking data to determine the best pricing strategies. This helps maximize revenue during peak seasons while maintaining occupancy during off-peak periods.

2. Understanding Customer Behavior

Data analytics helps hotels understand guest preferences, booking habits, and travel patterns. This information can be used to improve customer experience and increase guest loyalty.

3. Identifying Profitable Market Segments

Hotels receive bookings from different market segments such as corporate clients, travel agencies, and direct customers. By analyzing revenue contribution from each segment, hotels can focus on the most profitable segments.

4. Seasonal Demand Forecasting

Historical data helps hotels predict future demand patterns. For example, certain months may experience higher booking volumes due to holidays or tourism seasons.

5. Operational Efficiency

Data analytics helps hotel managers optimize staffing levels, room allocation, and service delivery based on occupancy patterns.

6. Strategic Decision Making

Interactive dashboards allow hotel executives to quickly monitor performance indicators and take strategic actions when necessary.

In this project, Power BI is used to transform hotel booking data into visual dashboards that provide meaningful insights into hotel operations.

1.3 Problem Statement

Despite having access to large amounts of operational data, many hotels struggle to effectively analyze and utilize this information. Without proper data analysis tools, it becomes difficult for hotel managers to identify important business trends and make data-driven decisions.

Some common challenges faced by hotels include:

- Difficulty in identifying profitable booking channels
- Lack of understanding of seasonal demand patterns
- Inefficient pricing strategies
- Limited insight into customer behavior
- Poor visibility of revenue performance

These challenges can negatively affect the hotel's profitability and operational efficiency.

Therefore, there is a need for an analytical solution that can transform raw booking data into meaningful insights.

This project addresses these challenges by developing a **Power BI dashboard** that provides a comprehensive analysis of hotel revenue and booking data from 2018 to 2020.

1.4 Aim of the Study

The main aim of this project is to analyze hotel booking and revenue data using Power BI to gain insights into hotel performance, customer behavior, and revenue trends.

The project focuses on transforming raw hotel data into interactive visual dashboards that support data-driven decision-making.

1.5 Structure of the Report

This project report is organized into five chapters, each focusing on a different aspect of the study.

Chapter 1 – Introduction

Provides background information about the hospitality industry, the importance of data analytics, and the objectives of the study.

Chapter 2 – Objectives of the Study

Describes the goals and objectives that guide the research.

Chapter 3 – Research Methodology

Explains the data sources, tools used, and analytical techniques applied in the study.

Chapter 4 – Data Analysis and Dashboard Insights

Presents the Power BI dashboard analysis and interprets the results.

Chapter 5 – Findings, Conclusion, and Recommendations

Summarizes the key findings and provides suggestions for improving hotel performance.

Chapter 2

OBJECTIVES OF THE STUDY

Every research project is guided by a set of objectives that define the purpose and direction of the study. Objectives help the researcher focus on specific goals and ensure that the analysis is carried out in a structured and systematic manner.

In the hospitality industry, understanding hotel performance requires analyzing several factors such as revenue generation, booking patterns, customer behavior, and operational efficiency. By defining clear objectives, it becomes easier to interpret data and derive meaningful insights that can support managerial decision-making.

The primary purpose of this study is to analyze hotel booking and revenue data from **2018 to 2020** using Power BI in order to identify trends, patterns, and insights that can help improve hotel performance.

The study focuses on evaluating key performance indicators (KPIs) such as total revenue, average daily rate, total bookings, customer type distribution, and length of stay. By analyzing these metrics, the project aims to provide a comprehensive understanding of hotel operations and customer behavior.

2.1 Primary Objective

The main objective of this project is to analyze hotel booking and revenue data using Power BI in order to gain insights into the performance of hotels and understand factors that influence revenue generation.

The analysis aims to convert raw data into meaningful visual dashboards that help identify trends, patterns, and opportunities for improvement in hotel management.

2.2 Specific Objectives of the Study

To achieve the main objective, the study focuses on several specific objectives that cover different aspects of hotel operations.

2.2.1 To analyze total revenue generated by hotels

Revenue generation is one of the most important indicators of hotel performance. This objective focuses on evaluating the total revenue generated during the study period.

By analyzing revenue data across different years, the study aims to identify:

- Revenue growth or decline trends
- Seasonal revenue fluctuations
- Revenue contribution from different hotel types

Understanding revenue patterns helps hotel management develop effective pricing strategies and improve financial performance.

2.2.2 To compare revenue performance between different hotel types

The dataset used in this project includes two types of hotels:

- City Hotel
- Resort Hotel

Each type of hotel serves a different target market. City hotels usually cater to business travelers and short-term stays, while resort hotels are more popular among tourists and vacation travelers.

This objective aims to compare the revenue contribution of both hotel types and identify which type generates higher revenue. The comparison helps understand how location and customer preferences influence hotel profitability.

2.2.3 To analyze booking trends over time

Booking trends provide valuable insights into customer demand and hotel occupancy patterns. By studying booking trends over months, quarters, and years, hotel managers can identify peak seasons and periods of low demand.

This objective focuses on analyzing:

- Total bookings per year

- Monthly booking patterns
- Quarterly booking distribution

Understanding booking trends helps hotels prepare for high-demand periods and develop strategies to attract customers during slower seasons.

2.2.4 To analyze market segment contribution

Hotels receive bookings from different market segments, each representing a different source of customers. These segments may include:

- Online Travel Agencies (OTA)
- Offline Travel Agents
- Direct bookings
- Corporate clients
- Group bookings
- Complementary bookings

This objective aims to identify which market segments generate the highest number of bookings and the highest revenue. By understanding the contribution of each segment, hotel management can focus marketing efforts on the most profitable channels.

2.2.5 To analyze customer booking behavior

Customer behavior plays an important role in hotel performance. Understanding how customers book rooms, how often they return, and how long they stay helps hotels improve customer satisfaction and loyalty.

This objective focuses on analyzing:

- New vs. repeated guests
- Customer types (transient, group, contract, etc.)
- Average length of stay

By studying customer behavior, hotels can design better loyalty programs and improve guest experience.

2.2.6 To evaluate seasonal demand patterns

The hotel industry is highly influenced by seasonal demand. Certain months of the year may experience higher booking rates due to holidays, festivals, or tourism seasons.

This objective aims to analyze seasonal demand by studying:

- Monthly stay patterns
- Peak travel seasons
- Off-peak demand periods

Understanding seasonal trends helps hotel managers optimize pricing strategies and resource allocation.

2.2.7 To analyze operational insights

Operational insights provide information about how efficiently the hotel operates. This includes factors such as:

- Weekend vs. weekday revenue
- Total number of nights stayed by guests
- Special requests made by customers

Analyzing these operational metrics helps hotels improve service quality and operational efficiency.

2.3 Significance of the Objectives

The objectives of this study are important because they help identify key factors that influence hotel revenue and performance. By analyzing booking and revenue data, hotel managers can gain valuable insights that support better decision-making.

The findings from this study can help hotels:

- Improve revenue management strategies
- Identify profitable customer segments
- Optimize marketing channels

- Enhance customer satisfaction
- Improve operational efficiency

These insights can contribute to the overall growth and success of hotel businesses.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

Research methodology refers to the systematic process used to collect, analyze, and interpret data in order to achieve the objectives of a study. It provides a structured approach that ensures the reliability and validity of the research findings.

In this project, the research methodology focuses on analyzing hotel booking and revenue data using **Power BI**. The methodology includes data collection, data preparation, data transformation, and data visualization. The goal is to convert raw hotel booking data into meaningful insights that help understand revenue trends, booking patterns, and customer behavior.

The analysis process involves several steps, including data cleaning, data modeling, creation of calculated measures, and development of interactive dashboards. These steps help ensure that the data used in the analysis is accurate and suitable for generating reliable results.

3.2 Research Design

The research design used in this project is **descriptive and analytical in nature**.

Descriptive Research

Descriptive research focuses on describing the characteristics of a dataset and identifying patterns or trends. In this project, descriptive analysis is used to understand hotel booking patterns, revenue distribution, and customer behavior.

Analytical Research

Analytical research involves examining data to identify relationships and insights that can support decision-making. In this study, analytical techniques are applied to evaluate revenue performance, market segment contributions, and seasonal demand patterns.

By combining descriptive and analytical approaches, the project provides a comprehensive understanding of hotel operations.

3.3 Data Source

The data used in this project consists of hotel booking records collected from hotel management systems. The dataset includes information related to hotel reservations, customer details, booking channels, and revenue.

The dataset covers the time period from **July 2018 to August 2020** and includes information from two types of hotels:

- City Hotel
- Resort Hotel

The dataset contains several attributes that describe different aspects of hotel operations. Some of the important variables included in the dataset are:

- Hotel type
- Booking date
- Market segment
- Customer type
- Number of guests
- Length of stay
- Special requests
- Revenue generated

These variables help analyze various aspects of hotel performance such as revenue generation, booking behavior, and operational efficiency.

3.4 Data Collection Method

The data used for this analysis was obtained from hotel booking records that are typically stored in hotel reservation systems. These systems capture detailed information about every booking made by customers.

The collected data includes both numerical and categorical variables. Numerical variables include revenue, length of stay, and number of guests, while categorical variables include hotel type, customer type, and market segment.

After collecting the dataset, it was imported into **Power BI Desktop** for further processing and analysis.

3.5 Data Preparation and Cleaning

Before performing data analysis, it is important to prepare and clean the dataset to ensure accuracy and consistency. Raw data often contains missing values, duplicate records, or formatting errors that must be corrected.

The following data preparation steps were performed in this project:

1. Removing Duplicate Records

Duplicate entries in the dataset were identified and removed to ensure that each booking record was unique.

2. Handling Missing Values

Some records contained missing values. These values were either filled using appropriate methods or removed if they were not essential for analysis.

3. Data Formatting

Certain columns such as dates and numerical values were formatted properly to ensure compatibility with Power BI functions.

4. Data Transformation

Additional columns were created to simplify analysis. For example:

- Year and month extracted from booking date
- Total nights calculated from stay duration
- Revenue measures calculated using pricing data

These transformations helped improve the quality of the dataset and made it easier to perform analysis.

3.6 Tools and Technologies Used

The main tool used for this project is **Power BI**, a business intelligence platform developed by Microsoft. Power BI provides powerful features for data visualization and interactive reporting.

The following tools were used during the project:

Microsoft Power BI Desktop

Power BI Desktop was used to import the dataset, clean the data, create relationships between tables, and design interactive dashboards.

Power Query Editor

Power Query Editor was used to perform data cleaning and transformation tasks such as removing duplicates, filtering data, and creating calculated columns.

Data Modeling

Power BI's data modeling feature was used to create relationships between tables and develop measures for analyzing key performance indicators.

Data Visualization

Power BI provides a wide range of visualization tools such as bar charts, line charts, pie charts, maps, and KPI cards. These visualizations were used to create dashboards that display important insights.

3.7 Dashboard Development Process

The development of the Power BI dashboard involved several stages.

Stage 1: Data Import

The dataset was imported into Power BI using the **Get Data** feature. After importing the data, it was examined to ensure that all fields were correctly recognized.

Stage 2: Data Cleaning and Transformation

Data cleaning was performed using Power Query Editor. Unnecessary columns were removed, missing values were handled, and data types were adjusted.

Stage 3: Data Modeling

Relationships between tables were created to enable accurate analysis. Calculated measures were developed to compute key metrics such as total revenue, average daily rate, and total bookings.

Stage 4: Visualization

Various visualizations were created to represent the data in a meaningful way. These include:

- Revenue trend charts
- Booking distribution charts
- Market segment analysis
- Customer behavior charts
- Operational insights charts

Stage 5: Dashboard Design

Multiple visualizations were combined into a dashboard layout to provide an overview of hotel performance. Filters and slicers were added to allow users to interact with the data and explore specific segments.

3.8 Key Performance Indicators (KPIs) Used

Several KPIs were used in the dashboard to measure hotel performance. These include:

Total Revenue
Represents the total income generated from hotel bookings during the study period.

Average Daily Rate (ADR)
Represents the average revenue earned per occupied room.

Total Bookings

Represents the total number of reservations made during the study period.

Average Length of Stay

Represents the average number of days guests stay at the hotel.

Total Guests

Represents the total number of people staying in the hotel during the study period.

These KPIs help evaluate the financial and operational performance of hotels.

3.9 Limitations of the Study

Although the study provides valuable insights, it has certain limitations:

1. The analysis is limited to the dataset covering the period from **2018 to 2020**.
2. External factors such as economic conditions, tourism trends, and global events are not considered.
3. The analysis focuses only on two hotel types: City Hotel and Resort Hotel.
4. The study does not include predictive modeling or forecasting techniques.

Despite these limitations, the analysis provides useful insights into hotel booking patterns and revenue performance.

CHAPTER 4

DATA ANALYSIS AND DASHBOARD INSIGHTS

4.1 Introduction

Data analysis is one of the most important stages of any research project. It involves examining raw data in order to identify meaningful patterns, trends, and insights. In this project, the hotel booking dataset was analyzed using **Power BI**, which allows interactive visualization and exploration of data.

The Power BI dashboard developed for this project provides insights into different aspects of hotel operations, including revenue generation, booking patterns, customer behavior, market segments, and operational performance.

The analysis focuses on several **Key Performance Indicators (KPIs)** such as:

- Total Revenue
- Average Daily Rate (ADR)
- Total Bookings
- Average Length of Stay
- Number of Guests

From the dashboard analysis, the following overall performance indicators were identified:

- **Total Revenue:** 30.5 million
- **Total Bookings:** 86.8K
- **Total Guests:** 168.7K
- **Average Daily Rate (ADR):** 100.8
- **Average Length of Stay:** 3.5 days

These metrics provide an overview of the hotel's financial and operational performance during the study period.

4.2 Executive Overview

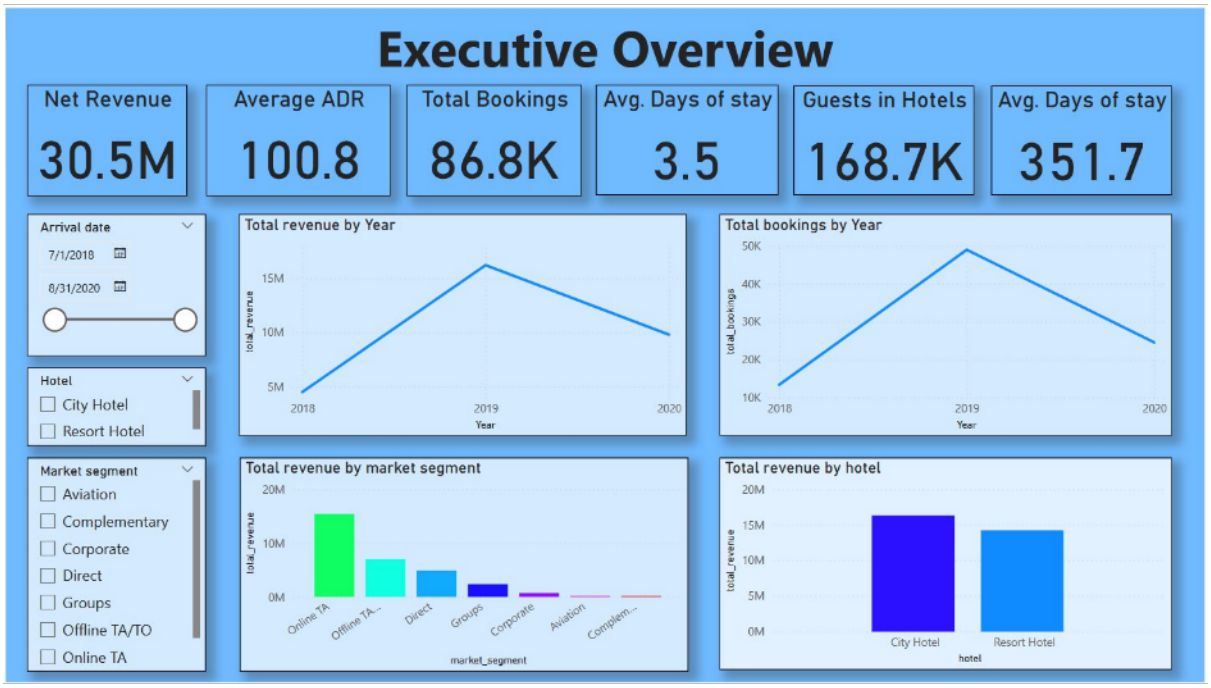
The executive overview provides a high-level summary of the hotel's overall performance.

Total Revenue Analysis

The total revenue generated during the study period is **30.5 million**, indicating strong financial performance. Revenue is generated through different booking channels and customer segments.

The analysis shows that revenue is influenced by factors such as:

- Hotel type
- Booking channel
- Seasonal demand
- Customer type



Understanding these factors helps hotel management optimize pricing strategies and increase profitability.

4.3 Revenue Analysis by Hotel Type

The dataset includes two types of hotels:

- City Hotel
- Resort Hotel

City Hotel

City hotels are typically located in urban areas and mainly serve business travelers, conference attendees, and short-term visitors.

These hotels tend to have:

- Higher occupancy rates during weekdays
- Shorter average stays
- More business-related bookings

Resort Hotel

Resort hotels are usually located in tourist destinations and cater mainly to vacation travelers.

Resort hotels generally experience:

- Higher demand during holidays and vacation seasons
- Longer average stays
- Higher weekend occupancy

Insight

The analysis shows that both hotel types contribute significantly to total revenue, but the distribution of bookings and customer types varies between them. Understanding these differences helps hotel managers design targeted marketing strategies.

4.4 Revenue Analysis by Market Segment

Market segments represent the different sources from which hotel bookings originate. The main market segments identified in the dataset include:

- Online Travel Agencies (OTA)
- Offline Travel Agents
- Direct bookings
- Corporate clients
- Group bookings
- Aviation
- Complementary bookings

Online Travel Agencies (OTA)

Online travel agencies represent one of the largest sources of bookings. Customers often prefer online platforms because they allow easy comparison of prices and services.

Offline Travel Agents

Traditional travel agents also contribute to hotel bookings, especially for organized travel packages.

Direct Bookings

Direct bookings occur when customers book rooms directly through the hotel's website or reservation desk. These bookings are usually more profitable because they avoid third-party commissions.

Corporate Bookings

Corporate clients often book rooms for business travel, meetings, and conferences.

Group Bookings

Group bookings include reservations made for events, tours, or conferences.

Insight

The analysis indicates that **online travel agencies contribute the highest number of bookings and revenue**. Hotels can use this information to strengthen partnerships with online booking platforms.

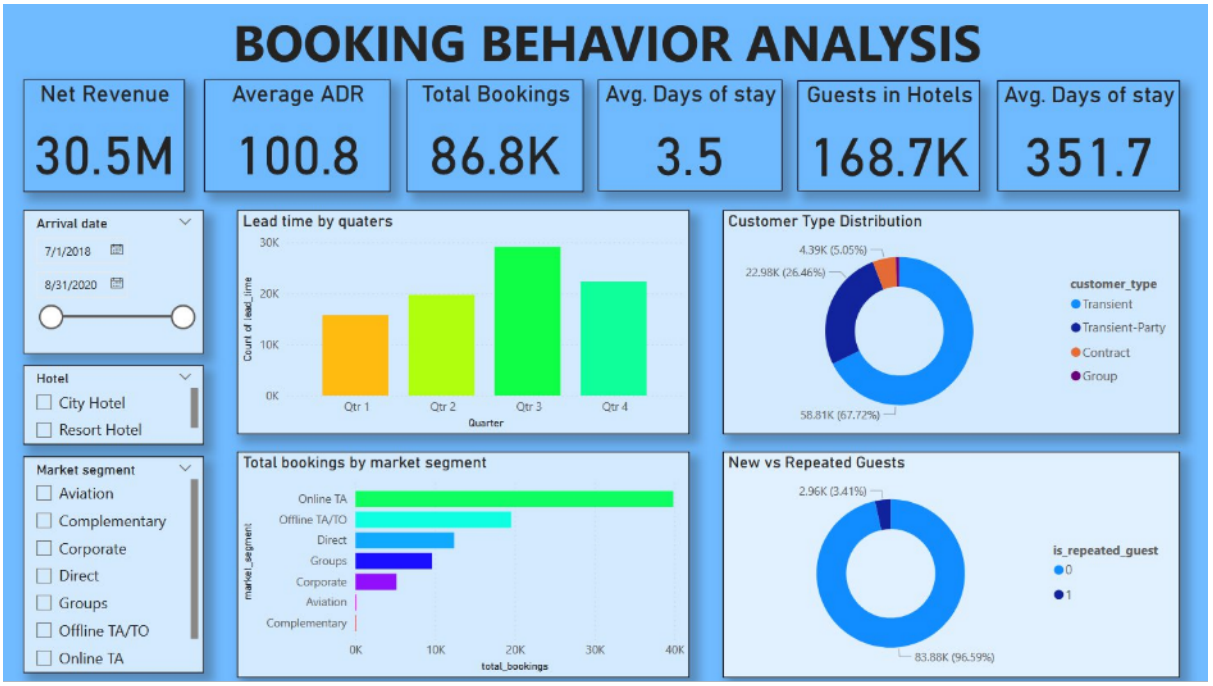
4.5 Booking Behaviour Analysis

Understanding customer booking behavior is essential for improving hotel services and marketing strategies.

New vs Repeated Guests

The dataset shows that the majority of hotel guests are **new customers**, while a smaller percentage are **repeat customers**.

- New Guests: approximately **96.59%**
- Repeated Guests: approximately **3.41%**



Insight

This indicates that hotels attract a large number of first-time visitors, but the repeat customer rate is relatively low. Increasing repeat customers through loyalty programs could help improve long-term revenue.

4.6 Customer Type Distribution

Customers are categorized into different types based on how they book their stay. The main customer types include:

- Transient
- Transient Party
- Contract
- Group

Transient Customers

These customers usually book individual rooms for short stays. They represent the largest portion of hotel bookings.

Transient Party

These customers travel in small groups but book rooms individually.

Contract Customers

Contract customers are usually business organizations that have agreements with hotels for regular bookings.

Group Customers

Group customers include large travel groups, conferences, or tour packages.

Insight

The analysis shows that **transient customers dominate the booking distribution**, indicating that most hotel stays are individual bookings rather than large group reservations.

4.7 Seasonal Booking Trends

The analysis also examines booking patterns across different months and quarters.

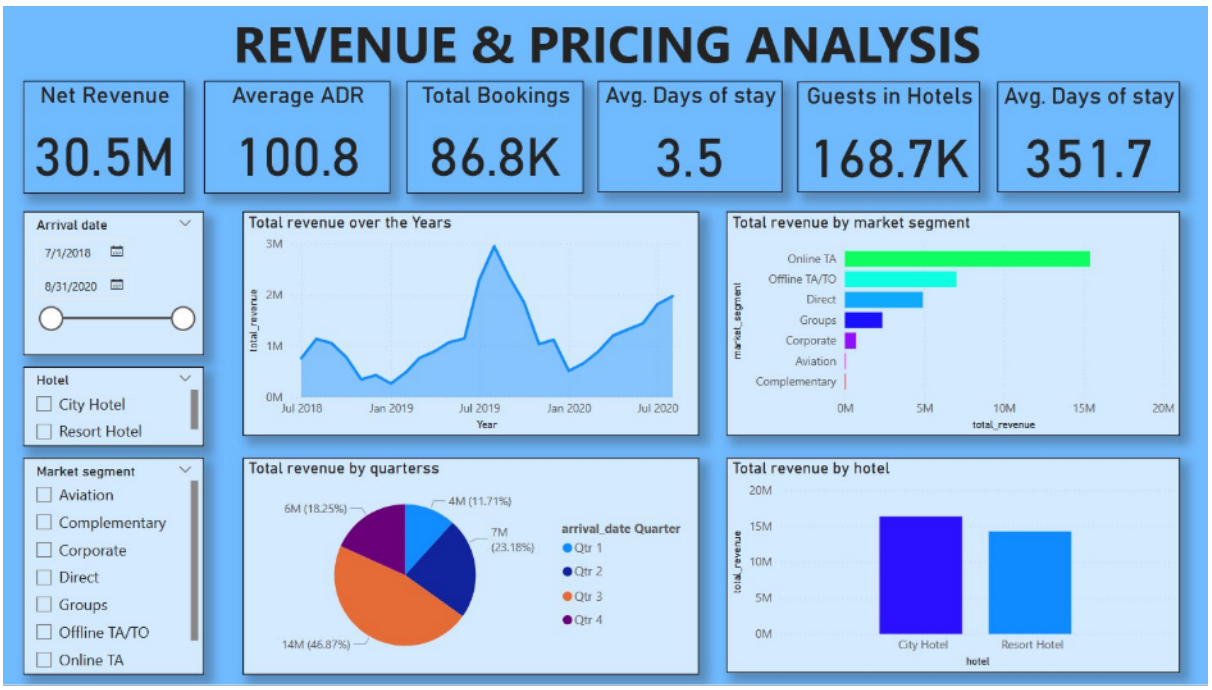
Quarterly Revenue Distribution

Revenue is distributed across four quarters of the year:

- Quarter 1
- Quarter 2

- Quarter 3
- Quarter 4

The data shows that **Quarter 3 generates the highest revenue**, indicating that it may correspond to peak travel seasons such as summer vacations.



Monthly Stay Patterns

The analysis of monthly stays shows variations in demand throughout the year. Certain months experience higher occupancy rates due to holidays and tourism seasons.

Insight

Understanding seasonal demand helps hotels adjust pricing strategies and marketing campaigns to maximize revenue during peak seasons.

4.8 Lead Time Analysis

Lead time refers to the number of days between when a booking is made and the actual arrival date.

The analysis shows that lead time varies across different quarters of the year.

Insight

Longer lead times usually indicate that customers plan their trips in advance, particularly for vacation travel. Shorter lead times are more common for business travel.

Hotels can use lead time analysis to predict demand and adjust room pricing accordingly.

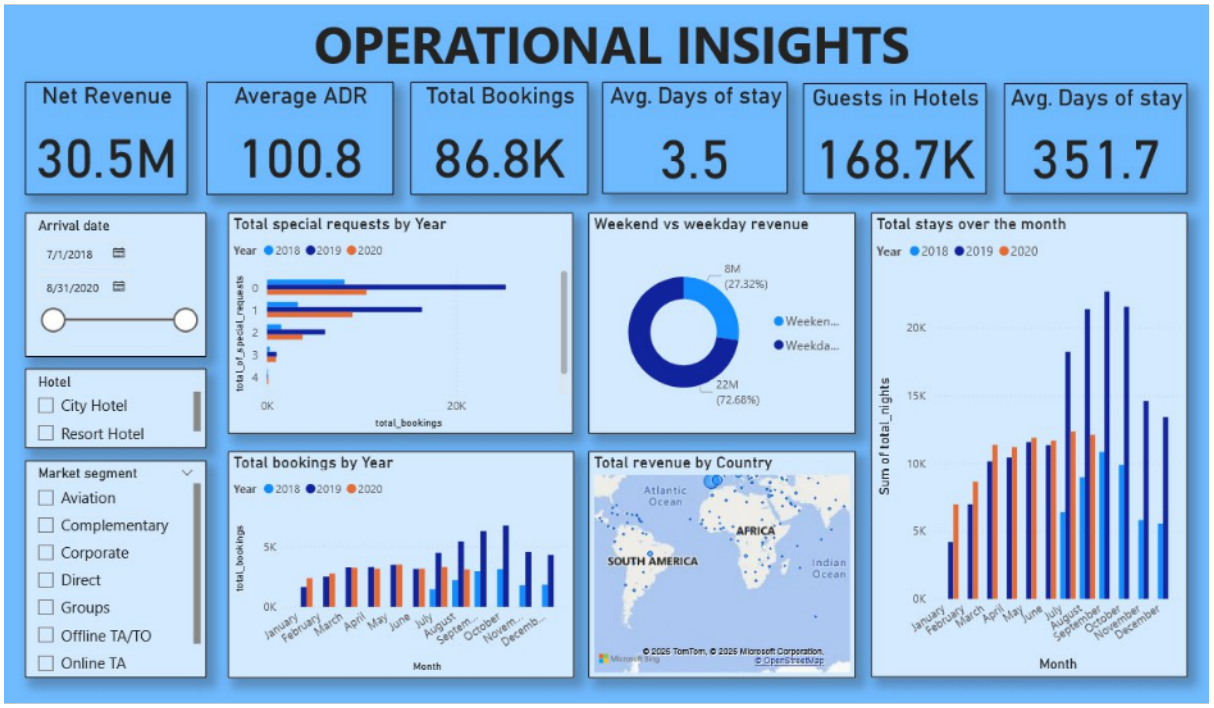
4.9 Operational Insights

Operational insights provide information about how hotel services are utilized.

Weekend vs Weekday Revenue

The analysis shows that:

- **Weekday revenue contributes around 72.68%**
- **Weekend revenue contributes around 27.32%**



Insight

This suggests that the hotels in the dataset receive more bookings during weekdays, likely due to business travelers.

Total Special Requests

Guests sometimes request special services such as:

- Extra beds
- Room preferences
- Early check-in
- Late check-out

The analysis shows that special requests increase gradually over the years, indicating increasing customer expectations for personalized services.

Total Stays Over the Months

The number of nights stayed by guests varies across different months.

Months with higher tourism activity show a larger number of total stays.

Insight

Hotels can use this information to prepare staffing schedules and resource allocation during peak periods.

4.10 Summary of Analysis

The Power BI dashboard provides a comprehensive view of hotel operations. The analysis highlights several important insights:

- Total revenue generated during the study period is **30.5 million**.
- Online travel agencies are the largest source of bookings.
- Most customers are first-time guests.
- Transient customers represent the majority of bookings.
- Quarter 3 generates the highest revenue.
- Weekday bookings contribute more revenue than weekend bookings.

These insights help hotel managers understand customer behavior, optimize pricing strategies, and improve operational efficiency.

CHAPTER 5

FINDINGS, CONCLUSION, AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the key findings obtained from the analysis of hotel booking and revenue data using the Power BI dashboard. The purpose of this chapter is to summarize the most important insights derived from the data analysis and provide conclusions based on the results.

The findings highlight important patterns related to hotel revenue, booking behavior, customer segments, and operational performance. Based on these insights, several recommendations are also provided to help improve hotel management strategies and increase overall profitability.

5.2 Major Findings of the Study

Based on the Power BI dashboard analysis, several key findings were identified regarding hotel performance between 2018 and 2020.

1. Strong Overall Revenue Performance

The analysis shows that the hotels generated a total revenue of **30.5 million** during the study period. This indicates that the hotel business maintained stable revenue performance across the years.

The revenue trend also shows that demand fluctuates across different months and quarters due to seasonal travel patterns.

2. High Contribution from Online Travel Agencies

One of the most significant findings is that **Online Travel Agencies (OTA)** generate the highest number of bookings and revenue compared to other market segments.

Online platforms allow customers to easily compare hotel prices and make reservations quickly. As a result, many travelers prefer using these platforms.

This highlights the importance of maintaining strong partnerships with online booking platforms.

3. City Hotels and Resort Hotels Contribute Differently to Revenue

The dataset includes two types of hotels:

- City Hotel
- Resort Hotel

City hotels generally attract business travelers and short-term visitors, while resort hotels mainly serve tourists and vacation travelers.

Both hotel types contribute significantly to total revenue, but their demand patterns vary depending on location and customer preferences.

4. Majority of Guests Are First-Time Visitors

The analysis shows that approximately **96.59% of guests are new customers**, while only about **3.41% are repeat guests**.

This indicates that the hotel attracts a large number of first-time visitors but has relatively low customer retention.

Increasing the number of repeat customers could help improve long-term revenue stability.

5. Transient Customers Dominate the Booking Distribution

Customer type analysis shows that **transient customers represent the largest group of hotel guests**. These customers typically book rooms individually for short stays.

Other customer types such as group bookings and contract bookings represent smaller portions of total reservations.

This suggests that hotels rely heavily on individual travelers rather than organized groups.

6. Quarter 3 Generates the Highest Revenue

The quarterly revenue analysis indicates that **Quarter 3 contributes the highest share of total revenue**.

This quarter likely corresponds to peak travel seasons such as summer holidays, when tourism activity increases significantly.

Hotels usually experience higher occupancy rates during this period.

7. Weekday Revenue Is Higher Than Weekend Revenue

Another important finding is that **weekday bookings contribute approximately 72.68% of total revenue**, while weekend bookings contribute about **27.32%**.

This suggests that business travel plays a major role in hotel occupancy, as business travelers typically stay during weekdays.

8. Average Length of Stay Is Relatively Short

The average length of stay for hotel guests is approximately **3.5 days**.

This indicates that most customers prefer short-term stays rather than long vacations.

Hotels may consider offering promotional packages for extended stays to increase revenue.

9. Increasing Number of Special Requests

The analysis also shows that the number of special requests made by guests increases over time.

This indicates that customers expect personalized services and better hospitality experiences.

Hotels need to focus on improving customer service to meet these expectations.

5.3 Conclusion

The hotel revenue analysis project provides valuable insights into the performance of hotels between 2018 and 2020. By using Power BI dashboards, large volumes of booking data were transformed into meaningful visualizations that help understand hotel operations more effectively.

The study highlights the importance of data analytics in the hospitality industry. Through the analysis of booking trends, revenue distribution, and customer behavior, hotel managers can make informed decisions that improve operational efficiency and profitability.

The results show that online booking platforms play a major role in generating hotel revenue. Additionally, the majority of hotel guests are first-time visitors, and most bookings are made by transient customers.

Seasonal trends also influence hotel performance, with certain quarters generating higher revenue due to increased tourism activity.

Overall, the use of Power BI dashboards enables hotel managers to monitor key performance indicators and quickly identify trends that impact business performance.

5.4 Recommendations

Based on the findings of this study, several recommendations can be made to improve hotel performance and revenue management.

1. Improve Customer Retention Strategies

Since most guests are first-time visitors, hotels should focus on improving customer loyalty programs.

Strategies may include:

- Loyalty reward programs
- Membership discounts
- Personalized customer services

Encouraging repeat visits can help increase long-term revenue.

2. Strengthen Online Booking Channels

Since online travel agencies generate a large share of bookings, hotels should maintain strong partnerships with these platforms.

Hotels should also improve their own websites and mobile booking systems to attract more direct bookings.

3. Offer Special Packages During Off-Peak Seasons

Seasonal demand variations affect hotel occupancy rates. To maintain steady bookings during off-peak seasons, hotels can introduce promotional packages such as:

- Discounted room rates
- Holiday packages
- Long-stay discounts

These strategies can help attract more customers during slow periods.

4. Improve Customer Experience

Since customers increasingly make special requests, hotels should focus on improving service quality.

Providing personalized services such as:

- Customized room preferences
- Flexible check-in and check-out
- Special amenities

can enhance guest satisfaction and encourage repeat visits.

5. Utilize Data Analytics for Strategic Decisions

Hotels should continue using data analytics tools like Power BI to monitor business performance and identify trends.

Regular data analysis can help hotel managers:

- Optimize pricing strategies
- Improve marketing campaigns
- Predict future demand

5.5 Future Scope of the Study

Although this study provides valuable insights, there are opportunities for further research and improvement.

Future studies could include:

- Predictive analytics for demand forecasting
- Machine learning models for dynamic pricing
- Customer sentiment analysis using reviews
- Comparison with competitor hotel data

These advanced analytical techniques could provide deeper insights and further improve hotel revenue management strategies.